

Clinician review packet: cancer symptom and lab navigator

Prepared by Palash Rakshit, Oncovision project (github.com/palashraks-afk/OncoVision). Live prototype: oncovisionai.vercel.app/navigator. Estimated time: about one hour.

What this is. A patient-facing prototype that applies published referral criteria to a person's symptoms and lab results and explains why. It has no clinical content of its own: every rule below was transcribed from a guideline by a student with the help of an AI assistant, and **no clinician has reviewed any of it**. The tool says so to every user. Your review is what would let it be called reviewed.

What is being asked

1. Tick one box per rule: matches the current guideline, needs a change, or remove.
2. Give a view on the eight design questions below.
3. Check the five screening items against the current US Preventive Services Task Force text.
4. Say whether you are willing to be named as reviewer, and whether you wish to be an author. Neither is assumed.

Eight questions for you

1. **Version.** The rules are transcribed from the 2015 edition of NICE NG12. NICE has since updated it (secondary sources report August and October 2023, including a quantitative faecal immunochemical test, FIT, as an early step in the bowel pathway). Please compare each rule with the current version.
Your view: _____
2. **US relevance.** The audience is likely to include US users. Should the symptom rules be shown alongside, replaced by, or reconciled with US guidance?
Your view: _____
3. **Platelet gating (lung).** NG12 lists a high platelet count as a trigger for considering a chest X-ray. Applied literally to symptom-free adults over 40 it alerted 3.7% of a US sample. The rule file requires at least one symptom as well. Do you agree?
Your view: _____
4. **Lab-only alerts.** With no symptoms, the anaemia and iron-deficiency rules alone alert 16 to 27% of people aged 80+ and about 14% of women aged 40 to 49 (likely menstrual iron loss). Should lab-only alerts be shown to symptom-free people at all, or only with a symptom?
Your view: _____
5. **Wording.** Is every action sentence safe, accurate and appropriately cautious? The tool never says a person has cancer.
Your view: _____
6. **Tiers.** Are the three levels (see a doctor soon, worth raising, mention) a fair translation of the guideline's timeframes and its 'recommend' versus 'consider'?
Your view: _____

7. **Missing rules.** Are any commonly important presentations missing, for example for children, for thyroid, or for head and neck?

Your view: _____

8. **Safety text.** Is the emergency wording adequate? "Seek urgent care now if you cough up or vomit a lot of blood. Do the same for severe chest pain, trouble breathing, or black or bloody stools with faintness."

Your view: _____

The 34 navigator rules

lung_haemoptysis Lung Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

aged 40 or over AND coughing up blood

OR a chest x-ray report that suggests lung cancer

What the tool tells the person: Ask your doctor for a specialist appointment within about 2 weeks to check for lung cancer.

Source page cited: NG12 2015, lung and pleural cancers, p.45

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

lung_xray_symptoms Lung Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

aged 40 or over AND 2 or more of: a cough that has no known cause; shortness of breath with no known cause; chest pain with no known cause; weight loss with no known reason; loss of appetite with no known reason; tiredness or fatigue with no known cause

OR aged 40 or over AND have ever smoked AND any one of: a cough that has no known cause; shortness of breath with no known cause; chest pain with no known cause; weight loss with no known reason; loss of appetite with no known reason; tiredness or fatigue with no known cause

What the tool tells the person: Ask for a chest X-ray within about 2 weeks to check for lung cancer.

Source page cited: NG12 2015, lung and pleural cancers, p.45

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

lung_xray_consider Lung Cancer Worth raising · within 2 weeks · guideline strength: consider

Criteria in the file:

aged 40 or over AND chest infections that keep coming back or will not clear

OR aged 40 or over AND fingertips that have become wider and rounded, nails curving over

OR aged 40 or over AND swollen glands above the collarbone, or in the neck that do not go away

OR aged 40 or over AND a high platelet count AND at least one symptom with no known cause

What the tool tells the person: Ask your doctor if a chest X-ray within about 2 weeks is worth doing.

Source page cited: NG12 2015, lung and pleural cancers, p.45

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

meso_xray_symptoms Mesothelioma See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

aged 40 or over AND exposed to asbestos AND any one of: a cough that has no known cause; shortness of breath with no known cause; chest pain with no known cause; weight loss with no known reason; loss of appetite with no known reason; tiredness or fatigue with no known cause

OR a chest x-ray report that suggests mesothelioma

What the tool tells the person: Ask for a chest X-ray within about 2 weeks. Doctors check for mesothelioma when these symptoms come with asbestos exposure.

Source page cited: NG12 2015, lung and pleural cancers, p.49

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

upper_gi_urgent Oesophageal Or Stomach Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

trouble swallowing

OR aged 55 or over AND weight loss with no known reason AND any one of: pain in the upper tummy; heartburn or acid reflux; indigestion

What the tool tells the person: Ask for a camera test of the gullet and stomach (an endoscopy) within about 2 weeks.

Source page cited: NG12 2015, oesophageal and stomach cancers, pp.62 and 84

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

upper_gi_haematemesis Oesophageal Or Stomach Cancer Mention at a routine visit · routine / non-urgent · guideline strength: consider

Criteria in the file:

vomiting blood

What the tool tells the person: Tell a doctor. A non-urgent endoscopy may be considered. Vomiting a lot of blood is an emergency: seek care now.

Source page cited: NG12 2015, oesophageal and stomach cancers, pp.62 and 84

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

upper_gi_consider Oesophageal Or Stomach Cancer Mention at a routine visit · routine / non-urgent · guideline strength: consider

Criteria in the file:

aged 55 or over AND indigestion that does not improve with treatment

OR aged 55 or over AND pain in the upper tummy AND low haemoglobin (anaemia)

OR aged 55 or over AND a high platelet count AND any one of: nausea with no known cause; vomiting with no known cause; weight loss with no known reason; heartburn or acid reflux; indigestion; pain in the upper tummy

OR aged 55 or over AND any one of: nausea with no known cause; vomiting with no known cause AND any one of: weight loss with no known reason; heartburn or acid reflux; indigestion; pain in the upper tummy

What the tool tells the person: Ask whether a non-urgent endoscopy is worth doing.

Source page cited: NG12 2015, oesophageal and stomach cancers, pp.62 and 84

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

pancreas_jaundice Pancreatic Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

aged 40 or over AND yellow skin or yellow whites of the eyes

What the tool tells the person: Ask your doctor for a specialist appointment within about 2 weeks.

Source page cited: NG12 2015, pancreatic cancer, p.71

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

pancreas_scan Pancreatic Cancer Worth raising · within 2 weeks · guideline strength: consider

Criteria in the file:

aged 60 or over AND weight loss with no known reason AND any one of: diarrhoea with no known cause; back pain with no known cause; tummy pain with no known cause; nausea with no known cause; vomiting with no known cause; constipation with no known cause; diabetes diagnosed recently

What the tool tells the person: Ask whether an urgent scan (CT, or ultrasound if CT is not available) is worth doing.

Source page cited: NG12 2015, pancreatic cancer, p.71

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

colorectal_urgent Bowel Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

aged 40 or over AND weight loss with no known reason AND tummy pain with no known cause

OR aged 50 or over AND bleeding from the back passage

OR aged 60 or over AND iron-deficiency anaemia

OR aged 60 or over AND a lasting change in your bowel habit

OR a stool test that found hidden blood

What the tool tells the person: Ask your doctor for a specialist appointment within about 2 weeks to check for bowel cancer.

Source page cited: NG12 2015, colorectal cancer, p.137

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

colorectal_mass Bowel Cancer Worth raising · within 2 weeks · guideline strength: consider

Criteria in the file:

a lump you can feel in your tummy

OR a lump you can feel in the rectum or back passage

What the tool tells the person: Ask your doctor if you need a specialist appointment within about 2 weeks.

Source page cited: NG12 2015, colorectal cancer, p.137

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

colorectal_young_bleeding Bowel Cancer Worth raising · within 2 weeks · guideline strength: consider

Criteria in the file:

under 50 AND bleeding from the back passage AND any one of: tummy pain with no known cause; a lasting change in your bowel habit; weight loss with no known reason

OR under 50 AND bleeding from the back passage AND iron-deficiency anaemia

What the tool tells the person: Ask your doctor if you need a specialist appointment within about 2 weeks.

Source page cited: NG12 2015, colorectal cancer, p.137

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

colorectal_stool_test Bowel Cancer Worth raising · primary-care test · guideline strength: recommended

Criteria in the file:

no bleeding from the back passage AND aged 50 or over AND any one of: tummy pain with no known cause; weight loss with no known reason

OR no bleeding from the back passage AND under 60 AND a lasting change in your bowel habit

OR no bleeding from the back passage AND under 60 AND iron-deficiency anaemia

OR no bleeding from the back passage AND aged 60 or over AND low haemoglobin (anaemia)

What the tool tells the person: Ask for a stool test for hidden blood.

Source page cited: NG12 2015, colorectal cancer, p.137

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

anal_mass Anal Cancer Worth raising · within 2 weeks · guideline strength: consider

Criteria in the file:

a lump or sore (ulcer) at the anus that is not explained

What the tool tells the person: Ask your doctor if you need a specialist appointment within about 2 weeks.

Source page cited: NG12 2015, anal cancer, p.144

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

breast_urgent Breast Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

aged 30 or over AND a breast lump with no known cause

OR aged 50 or over AND discharge, pulling in, or another change in one nipple only

What the tool tells the person: A doctor should arrange an urgent breast clinic appointment (within about 2 weeks).

Source page cited: NG12 2015, breast cancer, p.155

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

breast_consider Breast Cancer Worth raising · within 2 weeks · guideline strength: consider

Criteria in the file:

skin changes on a breast that look unusual

OR aged 30 or over AND a lump in the armpit with no known cause

What the tool tells the person: Ask whether an urgent breast clinic appointment (within about 2 weeks) is needed.

Source page cited: NG12 2015, breast cancer, p.155

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

breast_young_lump Breast Cancer Mention at a routine visit · routine / non-urgent · guideline strength: consider

Criteria in the file:

under 30 AND a breast lump with no known cause

What the tool tells the person: Ask for a non-urgent referral to have the lump checked.

Source page cited: NG12 2015, breast cancer, p.155

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

ovarian_symptoms Ovarian Cancer Worth raising · primary-care test · guideline strength: recommended

Criteria in the file:

female AND any of these on 12 or more days a month: a swollen, bloated tummy that keeps coming back; feeling full quickly when eating; loss of appetite with no known reason; pain in the pelvis or tummy; needing to pass urine urgently or more often

What the tool tells the person: Ask for a CA-125 blood test. These symptoms, especially after 50 and on more than about 12 days a month, are the ones doctors test for ovarian cancer.

Source page cited: NG12 (ovarian cancer recommendations dated 2011), p.159

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

ovarian_ibs Ovarian Cancer Worth raising · primary-care test · guideline strength: recommended

Criteria in the file:

female AND aged 50 or over AND bowel symptoms that look like irritable bowel syndrome, starting in the last 12 months

What the tool tells the person: Ask for tests for ovarian cancer. Irritable bowel syndrome rarely appears for the first time at this age.

Source page cited: NG12 (ovarian cancer recommendations dated 2011), p.159

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

ovarian_ca125 Ovarian Cancer Worth raising · primary-care test · guideline strength: recommended

Criteria in the file:

female AND CA-125 of 35 or above AND any one of: a swollen, bloated tummy that keeps coming back; feeling full quickly when eating; loss of appetite with no known reason; pain in the pelvis or tummy; needing to pass urine urgently or more often

What the tool tells the person: A CA-125 at or above 35 with these symptoms means an ultrasound of the abdomen and pelvis should be arranged.

Source page cited: NG12 (ovarian cancer recommendations dated 2011), p.159

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

ovarian_other Ovarian Cancer Worth raising · primary-care test · guideline strength: consider

Criteria in the file:

female AND any one of: weight loss with no known reason; tiredness or fatigue with no known cause; a lasting change in your bowel habit

What the tool tells the person: Ask whether tests for ovarian cancer are worth doing, alongside looking for other causes.

Source page cited: NG12 (ovarian cancer recommendations dated 2011), p.159

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

prostate_psa Prostate Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

male AND a psa result your report marks as above the range for your age

What the tool tells the person: Ask your doctor for a specialist appointment within about 2 weeks.

Source page cited: NG12 2015, prostate cancer, p.178

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

prostate_symptoms Prostate Cancer Mention at a routine visit · routine / non-urgent · guideline strength: consider

Criteria in the file:

male AND getting up at night to pass urine, frequency, hesitancy, urgency, or difficulty emptying

OR male AND erection problems

OR male AND blood you can see in your urine

What the tool tells the person: Ask about a PSA blood test and a prostate examination.

Source page cited: NG12 2015, prostate cancer, p.178

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

bladder_urgent Bladder Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

aged 45 or over AND blood you can see in your urine

OR aged 60 or over AND blood found on a urine test that you cannot see AND pain when passing urine

OR aged 60 or over AND blood found on a urine test that you cannot see AND a raised white blood cell count

What the tool tells the person: Ask your doctor for a specialist appointment within about 2 weeks.

Source page cited: NG12 2015, bladder cancer, p.189

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

bladder_uti Bladder Cancer Mention at a routine visit · routine / non-urgent · guideline strength: consider

Criteria in the file:

aged 60 or over AND urine infections that keep coming back or will not clear

What the tool tells the person: Ask for a non-urgent referral to look for a cause of repeated urine infections.

Source page cited: NG12 2015, bladder cancer, p.189

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

renal_haematuria Kidney Cancer See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

aged 45 or over AND blood you can see in your urine

What the tool tells the person: Ask your doctor for a specialist appointment within about 2 weeks.

Source page cited: NG12 2015, renal cancer, p.203

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

leukaemia_children_blood_count Leukaemia See a doctor soon · within 48 hours · guideline strength: recommended

Criteria in the file:

under 25 AND any one of: looking unusually pale; tiredness or fatigue with no known cause; fevers with no known cause; infections that keep coming back or will not clear, with no known reason; swollen glands in several places at once; bone pain with no injury to explain it; bruising with no injury to explain it; bleeding with no injury to explain it

What the tool tells the person: Ask for a blood count within about 2 days. In children and young people these symptoms are the ones doctors check a blood count for, to look for leukaemia.

Source page cited: NG12 2015, leukaemia, p.256

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

leukaemia_children_petechiae Leukaemia See a doctor soon · immediately · guideline strength: recommended

Criteria in the file:

under 25 AND tiny red or purple spots on the skin that do not fade when pressed

What the tool tells the person: Tiny red or purple spots that do not fade under pressure in a child or young person need to be seen by a specialist straight away. Seek urgent care now.

Source page cited: NG12 2015, leukaemia, p.256

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

leukaemia_adult_blood_count Leukaemia Worth raising · within 48 hours · guideline strength: consider

Criteria in the file:

aged 25 or over AND any one of: looking unusually pale; tiredness or fatigue with no known cause; fevers with no known cause; infections that keep coming back or will not clear, with no known reason; swollen glands in several places at once; bruising with no injury to explain it; bleeding with no injury to explain it; tiny red or purple spots on the skin that do not fade when pressed

What the tool tells the person: Ask whether a blood count within about 2 days is worth doing. Most people with tiredness or similar symptoms do not have leukaemia.

Source page cited: NG12 2015, leukaemia, p.256

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

lymphoma_adult Lymphoma Worth raising · within 2 weeks · guideline strength: consider

Criteria in the file:

aged 25 or over AND a swollen gland or glands (neck, armpit or groin) with no infection to explain it

What the tool tells the person: Ask your doctor if you need a specialist appointment within about 2 weeks. Fever, night sweats, breathlessness, itching, or weight loss make it more likely.

Source page cited: NG12 2015, non-Hodgkin and Hodgkin lymphoma, pp.274 and 280

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

lymphoma_children Lymphoma Worth raising · within 48 hours · guideline strength: consider

Criteria in the file:

under 25 AND a swollen gland or glands (neck, armpit or groin) with no infection to explain it

What the tool tells the person: Ask whether a specialist appointment within about 2 days is needed. Fever, night sweats, breathlessness, itching, or weight loss make it more likely.

Source page cited: NG12 2015, non-Hodgkin and Hodgkin lymphoma, pp.274 and 280

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

weight_loss Several Cancers See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

weight loss with no known reason

What the tool tells the person: Weight loss with no known reason can be a sign of several cancers. These include bowel, stomach, gullet, lung, prostate, pancreas, kidney and bladder cancer. It can also have other causes. Guidelines say a doctor should look for the cause soon.

Source page cited: NG12 2015, symptoms that are not specific to one cancer, p.347

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

appetite_loss Several Cancers See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

loss of appetite with no known reason

What the tool tells the person: Losing your appetite for no known reason can be a sign of several cancers. These include lung, gullet, stomach, bowel, pancreas, bladder and kidney cancer. It can also have other causes. Guidelines say a doctor should look for the cause soon.

Source page cited: NG12 2015, symptoms that are not specific to one cancer, p.347

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

melanoma_7point Melanoma See a doctor soon · within 2 weeks · guideline strength: recommended

Criteria in the file:

Self-scored checklist: The mole or spot has changed in size (2); Irregular shape (2); Irregular colour (2); Wider than 7 mm (1); Looks inflamed (1); Oozing or crusting (1); Itching or other change in sensation (1). Threshold 3 or more.

What the tool tells the person: A doctor should look at this spot urgently (within about 2 weeks). A self-assessment is not reliable: a clinician needs to examine it.

Source page cited: NG12 2015, skin cancers, p.220

☐ matches the current guideline ☐ needs a change ☐ remove Comment:

The five US screening items

Each applies only to average-risk adults. Wording was checked against the Task Force pages on 5 October 2026.

Cancer	Source cited	Page	Your view
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breast cancer	USPSTF, Breast Cancer: Screening, April 30 2024 (Grade B)	https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/breast-cancer-screening	☐ ok ☐ change
cervical cancer	USPSTF, Cervical Cancer: Screening, August 21 2018 (Grade A)	https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/cervical-cancer-screening	☐ ok ☐ change
bowel (colorectal) cancer	USPSTF, Colorectal Cancer: Screening, May 18 2021 (Grade A for 50 to 75, Grade B for 45 to 49, Grade C for 76 to 85)	https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/colorectal-cancer-screening	☐ ok ☐ change
lung cancer	USPSTF, Lung Cancer: Screening, March 9 2021 (Grade B)	https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/lung-cancer-screening	☐ ok ☐ change
prostate cancer	USPSTF, Prostate Cancer: Screening, May 8 2018 (Grade C for 55 to 69, Grade D for 70 and older)	https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/prostate-cancer-screening	☐ ok ☐ change

Sign-off

Reviewer name and role: _____ Date: _____ Signature: _____

Overall: ☐ rules acceptable as a research prototype ☐ acceptable after the changes marked ☐ not acceptable